Understanding Trust Toward Human vs. AI-Generated Health Information Through Behavioral and Physiological Sensing

Sun, X., Ma, R., Wei, S., Cesar, P., Bosch, J.A., & El Ali, A. (2026). Int. J. Human-Computer Studies, 209, 103714.

What They Studied

As AI-generated health content becomes harder to tell apart from human-written content, this paper asks: how do people trust health information differently based on who (or what) wrote it, and what label it carries? They also test whether physiological signals from the body can reveal how trust works beneath the surface.

Two Studies

Study 1 - Online Survey (N=142): Participants read health Q&A pairs where the actual source (human professional vs. LLM) and the disclosed label (AI vs. Human) were independently manipulated. Three types tested: general, symptom-related, and treatment-related health info.

Study 2 - Lab Study (N=40): Same setup but with sensors: eye-tracker (fixation, saccade, pupil dilation) + ECG (heart rate variability), EDA (skin conductance), and skin temperature via electrodes and Pt1000 sensor.

Key Findings

- **LLM content was trusted more** than human-written content, regardless of labeling
- **Human labels were trusted more** than AI labels, regardless of actual source - labels matter more than reality
- Type of health question (general vs. symptom vs. treatment) did not significantly affect trust
- Eye-tracking: AI-labeled info triggered more scanning (more saccades, shorter fixations); human-labeled info led to deeper reading (longer fixations, larger pupils)
- RMSSD (HRV) was higher for AI-labeled info, suggesting lower physiological arousal
- Skin temperature was significantly higher for both AI-labeled and LLM-sourced content

ML Prediction Results

73%	65%	$R^2 = 0.35$
Trust level (binary)	Source classification	Trust score (regression)

Gaze features alone achieved best trust classification. Combining gaze + physiology achieved best source classification. Models: Random Forest, AdaBoost, XGBoost with LOSO cross-validation and SHAP feature importance.

Design Recommendations

- **Transparency labels matter** - but poorly framed labels can erode trust rather than build it
- Build "disclosure-aware" interfaces that dynamically adjust labels based on user state (e.g., surface more attribution when the system detects low attention or high cognitive load)
- **Real-time physiological state estimation is feasible** - gaze and physiology can serve as implicit trust indicators alongside self-reports

Limitations

- Single time-point, lab setting - they call for longitudinal, ecological studies
- Signals analyzed mostly independently, not as inter-signal relationships
- Only tested with GPT-4 - may not generalize across LLMs
- Young, tech-literate participant sample - limited generalizability

> **Core Takeaway**
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> Trust in health information is shaped more by how it's labeled than by who actually wrote it. Physiological signals from wearable sensors can detect these trust states implicitly - opening the door for adaptive, trust-aware health systems.